

Sufficient adjustment sets in DAGs

STSCI / INFO / ILRST 3900: Causal Inference

Sep 29 2026

Learning goals for today

At the end of class, you will be able to:

1. Identify a sufficient adjustment set using the backdoor criterion
2. Assess whether selection bias may hold in a gathered sample

Logistics

- ▶ Quiz 2 Oct 8
- ▶ Project assignments

Big picture

- ▶ If we can find covariates that yield conditional exchangeability, we can estimate causal effects (even in observational data) using standardization or IPW
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- ▶ Given a graph and a conditioning set L , we can determine if conditional exchangeability holds or not
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 2. Check if each path is open or blocked conditional on L
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If we believe the causal DAG, then we have reason to believe the estimated causal effect

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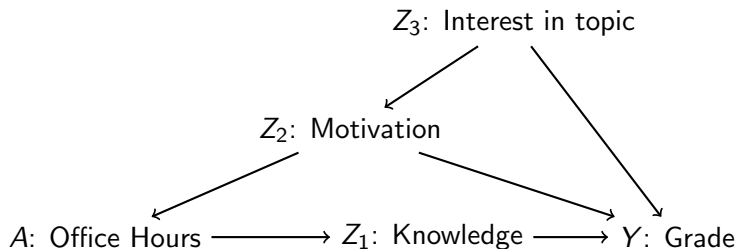
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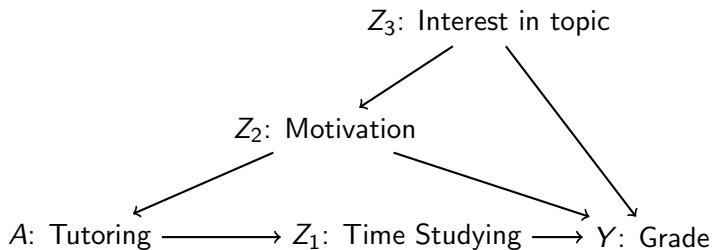
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Exercise



- What are the paths from A to Y ?
- When conditioning on $L = \{Z_1, Z_3\}$ are those paths open or blocked?
- When conditioning $L = \{Z_2\}$ are those paths open or blocked?

Exercise



Sufficient adjustment set

To conduct a causal analysis, we need to find a conditioning set of variables L which give us conditional exchangeability

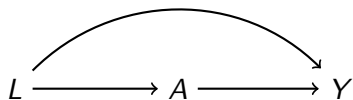
- ▶ Find a set of variables L that blocks all non-causal paths from A and Y
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Sufficient adjustment set

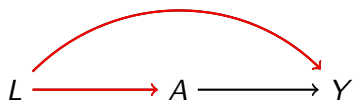
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- ▶ L is called **sufficient adjustment set**
- ▶ Various ways to do this, we will talk about one criterion

Backdoor criterion

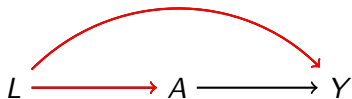


Backdoor criterion



Backdoor path starts with an edge pointing in to A and ends at Y

Backdoor criterion

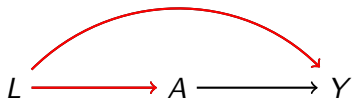


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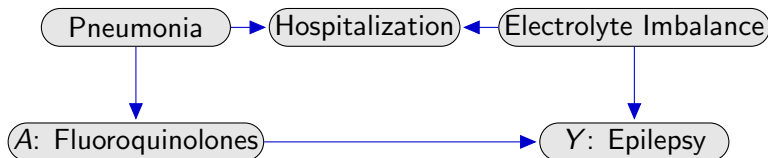
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Sets that satisfy the backdoor criterion are sufficient adjustment sets!

Exercise

Researchers may be interested in the effect of fluoroquinolones, a class of antibiotics, on epilepsy¹

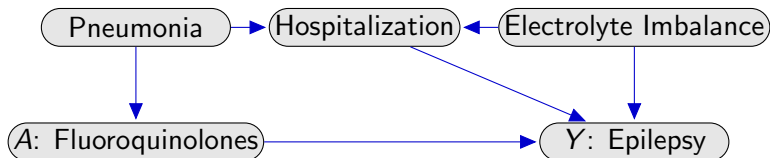


- Does a sufficient adjustment set exist? If so, what is it?

¹Example from “Using Causal Diagrams to Improve the Design and Interpretation of Medical Research” (Etminan et. al. 2020, Chest)

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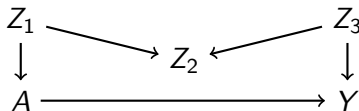
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Selection Bias

- ▶ Sufficient adjustment set to close backdoor paths

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- ▶ Does not always mean conditioning on more things



Selection Bias

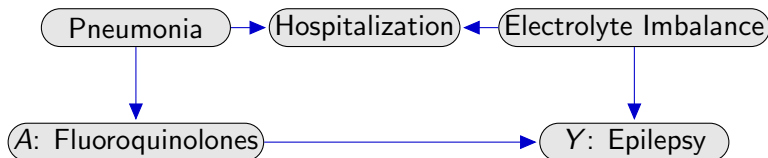
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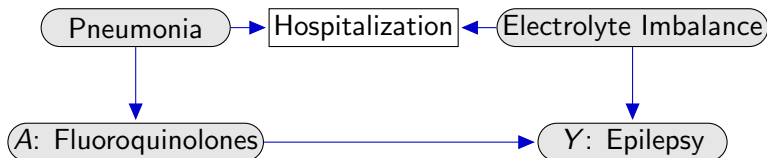


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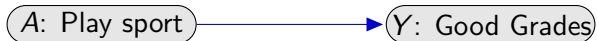
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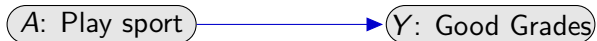


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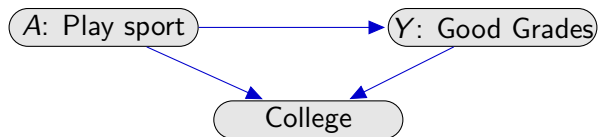
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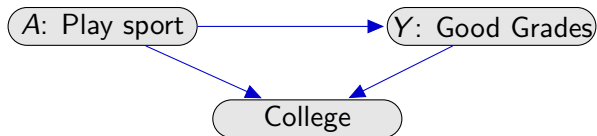
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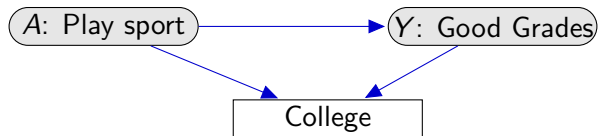


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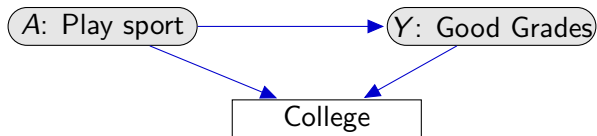


$$A \perp\!\!\!\perp Y^a$$

Selection Bias



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$$A \not\perp Y^a \mid \text{College}$$

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- ▶ May open non-causal paths

Takeaways

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 - ▶ If it does not exist, consider gathering more variables
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- ▶ Carefully consider the data gathering process
- ▶ Causal claims come from assumptions + data

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